

Recipe Sharing Platform with Meal Planner

Abstract and design proposal

TEAM ID 28 · SECTION S-10

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Problem Statement

Home cooks lack one reliable place to keep recipes, plan the week and build a shopping list.

1 Recipes are scattered

Notes, bookmarks and chats hold recipes in different places.

2 Planning is repetitive

Weekly menus are rebuilt manually each week.

3 Lists are hard to merge

Ingredients across several recipes must be combined by hand.

4 Nutrition is guesswork

Nutrition facts live in separate apps, if checked at all.

Existing System

Feature	Existing system	Proposed system
Recipe storage	Notes, bookmarks, chat messages	One searchable, categorised database
Meal planning	Done manually from memory	Weekly planner by day and meal
Grocery list	Written by hand, recipe by recipe	Auto-merged from the meal plan
Nutrition info	Separate apps or guesswork	Shown on each recipe via API
Limitation	Tools are separate and unlinked	Recipe, plan and list in one flow

Proposed System

1

Accounts

Register, log in, manage your own recipes.

2

Recipe sharing

Create, browse, search and rate recipes.

3

Meal planner

Fill a weekly plan from saved recipes.

4

Grocery list

Generated automatically from the plan.

5

Nutrition info

Calories, protein, carbs and fat per recipe.

6

Saved recipes

Keep favourites for quick reuse.

Objectives

- 1 Accounts to create, share, search and rate recipes
- 2 A weekly meal planner built from saved recipes
- 3 An auto-merged grocery list from the planned recipes
- 4 Nutrition information displayed on each recipe
- 5 A MongoDB data model of four linked collections
- 6 Secure login with JWT and bcrypt-hashed passwords

Scope



In scope

- Registered users with login
- Recipe creation, search and ratings
- Weekly meal planner
- Grocery list generation
- Nutrition info via external API



Out of scope (for now)

- Payments
- Delivery of groceries
- Native mobile apps
- Social features such as comments and follows

Technology Identification

R **React.js**

Single-page frontend with
React Router and Axios

N **Node.js**

Backend JavaScript runtime

E **Express.js**

REST API framework

M **MongoDB +
Mongoose**

Document database and
data models

C **Cloudinary**

Image hosting; only URLs
stored

N **Nutrition API**

Nutrition facts per recipe

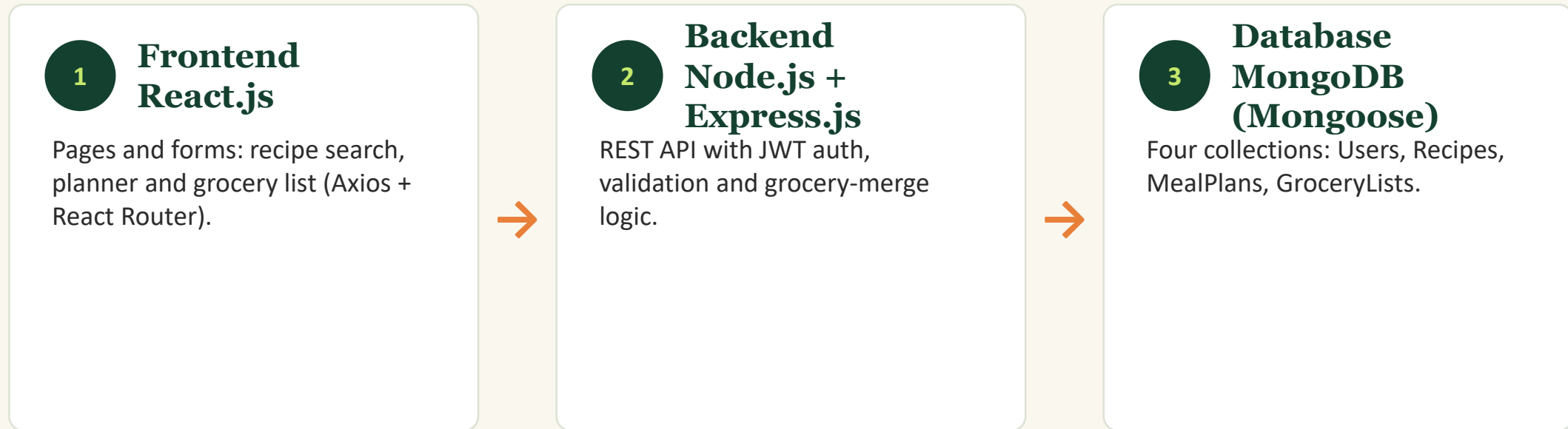
J **JWT + bcrypt**

Authentication and
password hashing

P **Postman · VS
Code · Git**

Testing, development,
version control

Basic Architecture



External services — Cloudinary (recipe images, only URLs stored) and a Nutrition API (calories, protein, carbs, fat) — are called only from the backend.

Collections & Relationships

1 Users

name, email (unique), password hash, savedRecipes[]

2 Recipes

author, title, category, ingredients[], ratings[], nutrition

3 MealPlans

userId, weekStart, entries[] (day, mealType, recipeId)

4 GroceryLists

userId, mealPlanId, items[] (name, qty, unit, checked)

User → Recipes: 1 to many (author), many to many (saved) · User → MealPlans: 1 to many · MealPlan → Recipes: many to many · MealPlan → GroceryList: 1 to 1

Team Details



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- Team ID 28 · Section S-10
- Guide: [to be added]
- Course: 25CS1302E — Database Systems Engineering and Distributed Backend Development
- Trimester 4, 2026–2027 · Stack: React · Node.js · Express.js · MongoDB

Abstract Summary

The Recipe Sharing Platform with Meal Planner brings recipes, weekly meal plans and shopping lists into one web application. Users share and rate recipes, plan their week from saved recipes, and get a grocery list merged automatically, with nutrition information alongside.

3-Tier

Architecture

4

Collections

REST

API

JWT

Authentication

Abstract submitted · Ready for Review 1 evaluation